

Analysis of Carl Zimmer's Personal Genome

Biomedical Data Science - Final Project: Group 6



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Identifying the top 10 genes on chr22 with the highest mutational burden.

1. VCF Filtering:

1. Extracted variants specific to **chr22** from Carl Zimmer's germline SNPs.

2. Gene Annotation:

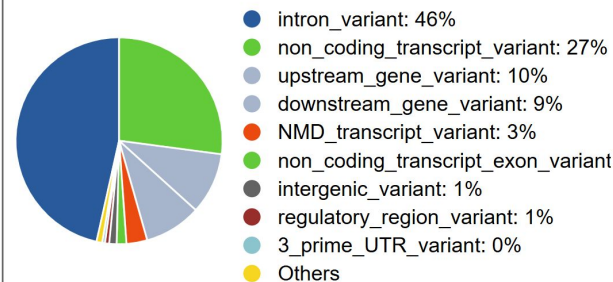
1. Used **Ensembl Variant Effect Predictor (VEP)** to annotate each variant with gene name and symbol.

3. Variant Counting:

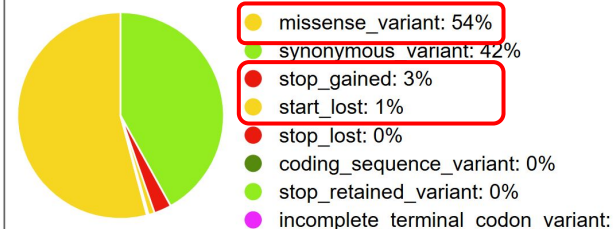
1. Counted number of variants/gene.
2. Selected **top 10 genes** based on highest variant counts.

Category	Count
Variants processed	45592
Variants filtered out	0
Novel / existing variants	43083 (94.5) / 2509 (5.5)
Overlapped genes	1613
Overlapped transcripts	7665
Overlapped regulatory features	1922

Consequences (all)



Coding consequences



Prioritizing genes not just by the number of variants, but by how severe those variants are likely to be

- 1. Impact Categorization** – Classified each variant into HIGH, MODERATE, LOW, or MODIFIER impact levels.
- 2. Scoring System** – Assigned a weighted score (example: HIGH = 3, MODERATE = 2, LOW = 1, MODIFIER = 0.1) to compute an overall IMPACT_SCORE per gene.
- 3. Ranking** – Top 10 genes were selected based on their cumulative IMPACT_SCORE.

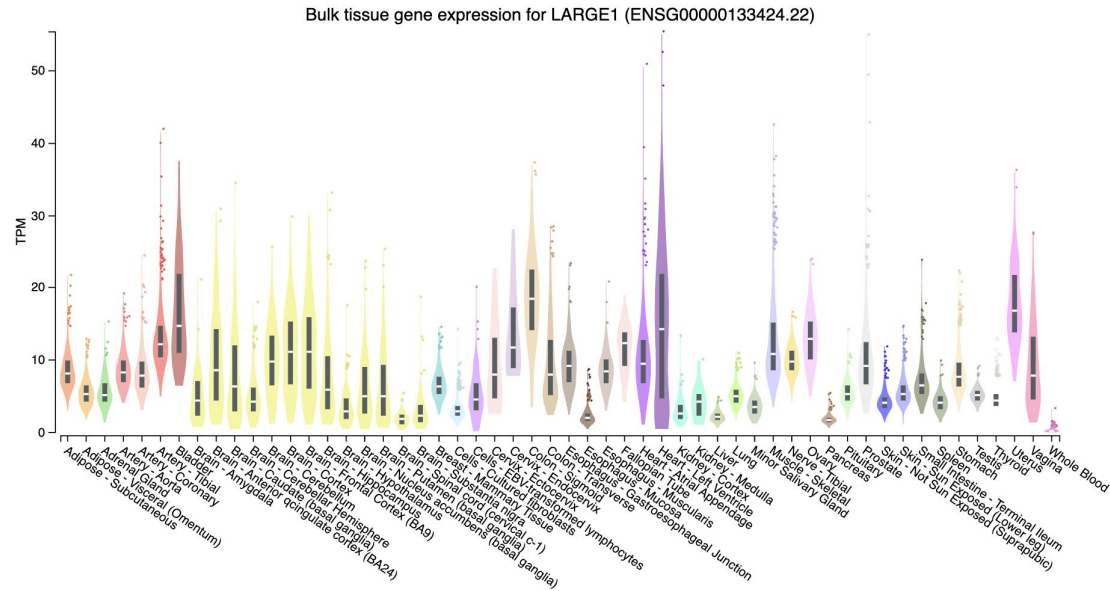
Top 10 Genes with Highest Mutational Burden on Chromosome 22

Gene ID	Symbol	Variant Count	Rank
ENSG00000133424	LARGE1	883	1
ENSG00000185666	SYN3	868	2
ENSG00000219438	TAF15	865	3
ENSG00000054611	TBC1D22A	848	4
ENSG00000244625	MIATNB	636	5
ENSG00000188511	MIR3667HG	590	6
ENSG00000166862	CACNG2	508	7
ENSG00000100154	TTC28	499	8
ENSG00000186976	EFCAB6	462	9
ENSG00000286592	LINC02885	459	10

Top 10 Genes by Variant Impact Score

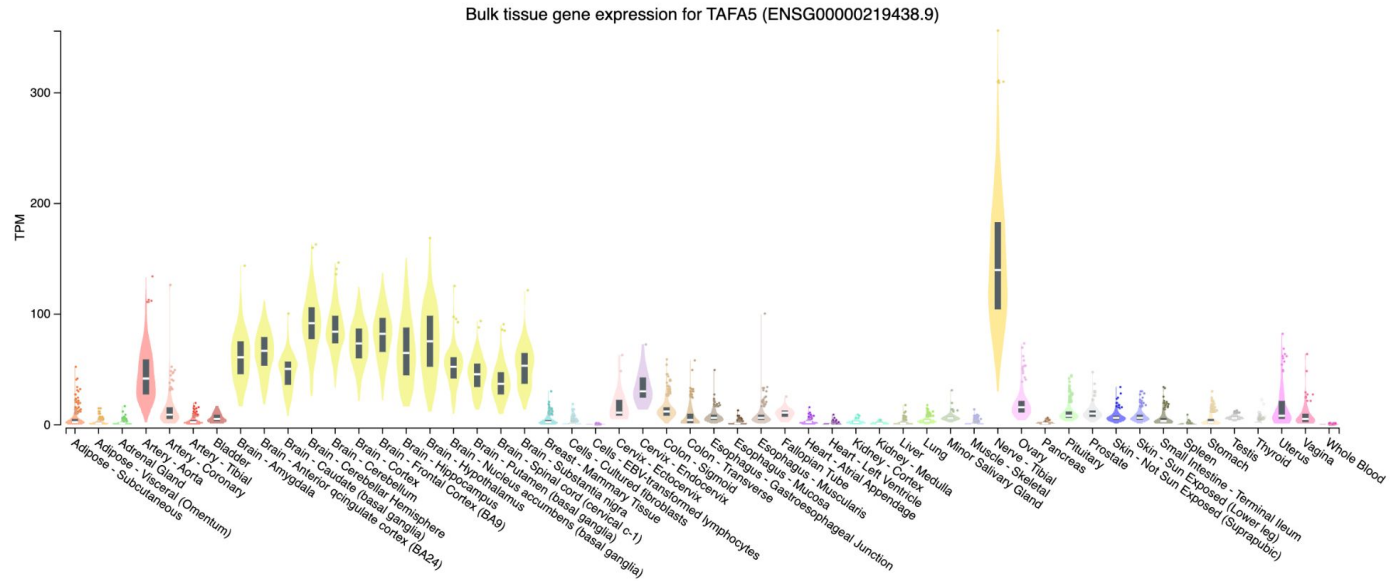
Gene Symbol	High	Moderate	Low	Modifier	Impact Score
BRD1	1	8	10	119	29
PLXNB2	4	5	5	74	27
DENND6B	2	5	9	57	25
NUP50	1	7	5	110	22
CACNG2	1	7	4	496	21
CECR2	1	4	10	364	21
EFCAB6	0	7	6	449	20
CABIN1	0	8	2	206	18
MNI	0	7	4	117	18
SERPIND1	2	5	2	40	18

1) LARGE1



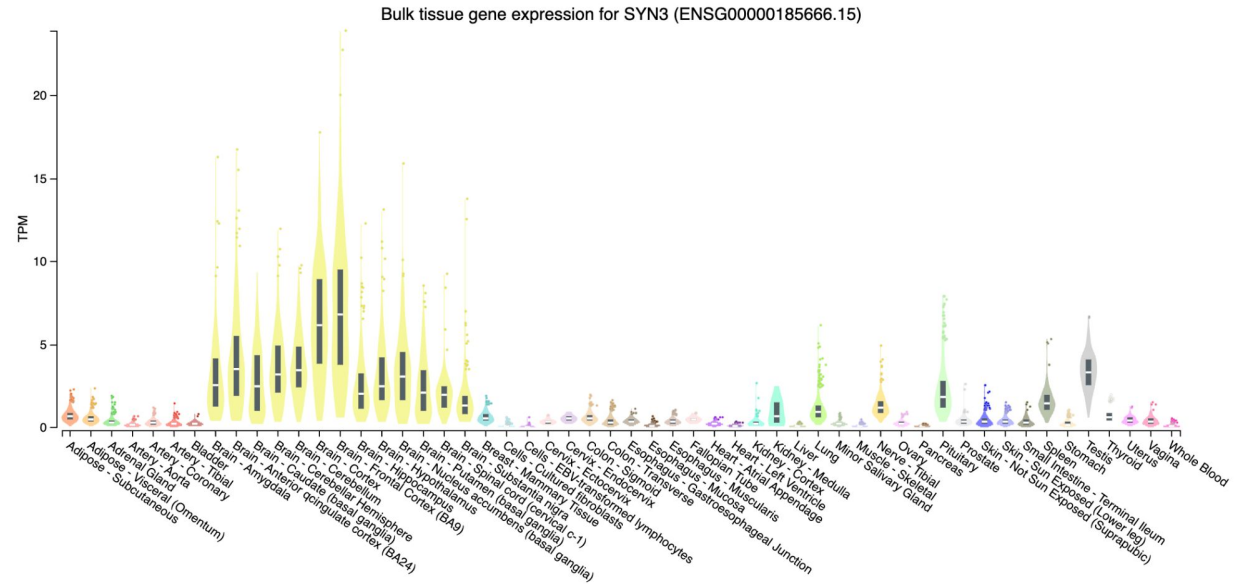
- **GTEx tissue expression:** highest in bladder, sigmoid colon, left ventricle, and uterus → high proportions of muscle tissue
- LARGE1 is a glycotransferase that generates matriglycan, which links the cytoskeleton and extracellular matrix
- Required for proper muscle function, mutations linked to congenital muscular dystrophy
- Can present with brain abnormalities

2) TAFA5



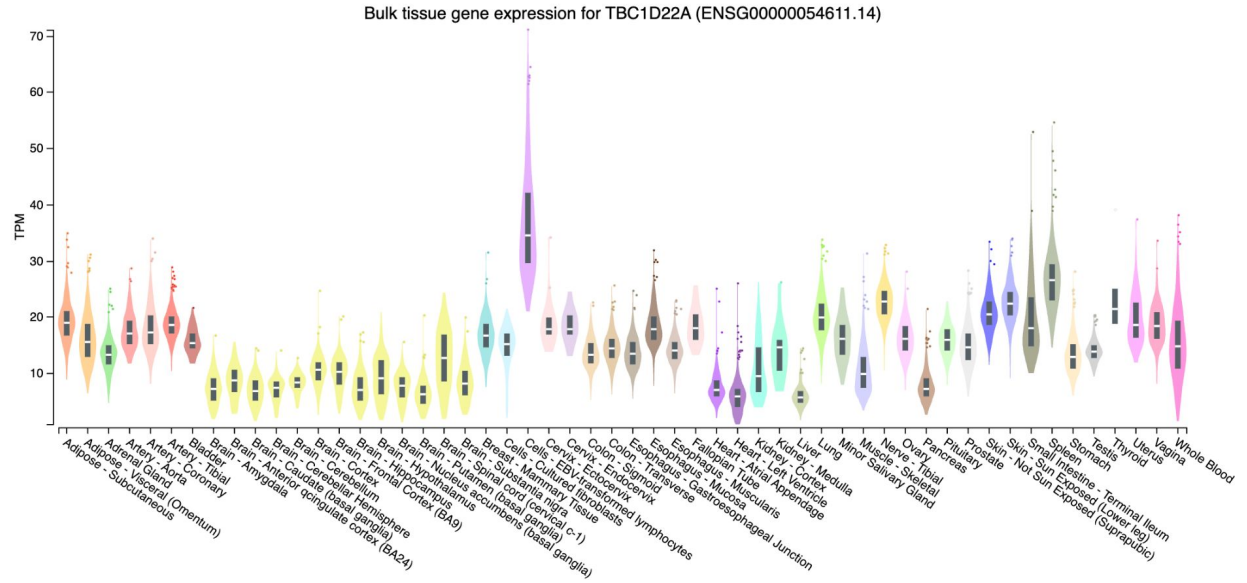
- **GTEx tissue expression:** high expression in all brain tissue
- TAFA5 is found in areas of inflammatory activation in the brain, triggering production of astrocytes after CNS injury
- Produced during neurogenesis of neural stem cells
- Genetic depletion of Tafa5 results in depressive behaviors and decreased hippocampus activity

3) SYN3



- **GTEx tissue expression:** high expression in all brain tissues, most notably the cerebral cortex
- SYN3 is a neuronal phosphoprotein that plays a significant role in synaptic transmission, neuronal plasticity, axon growth, and synapse formation
- Genetic variants for this gene have been associated with various brain disorders, such as epilepsy, schizophrenia, and bipolar disorder

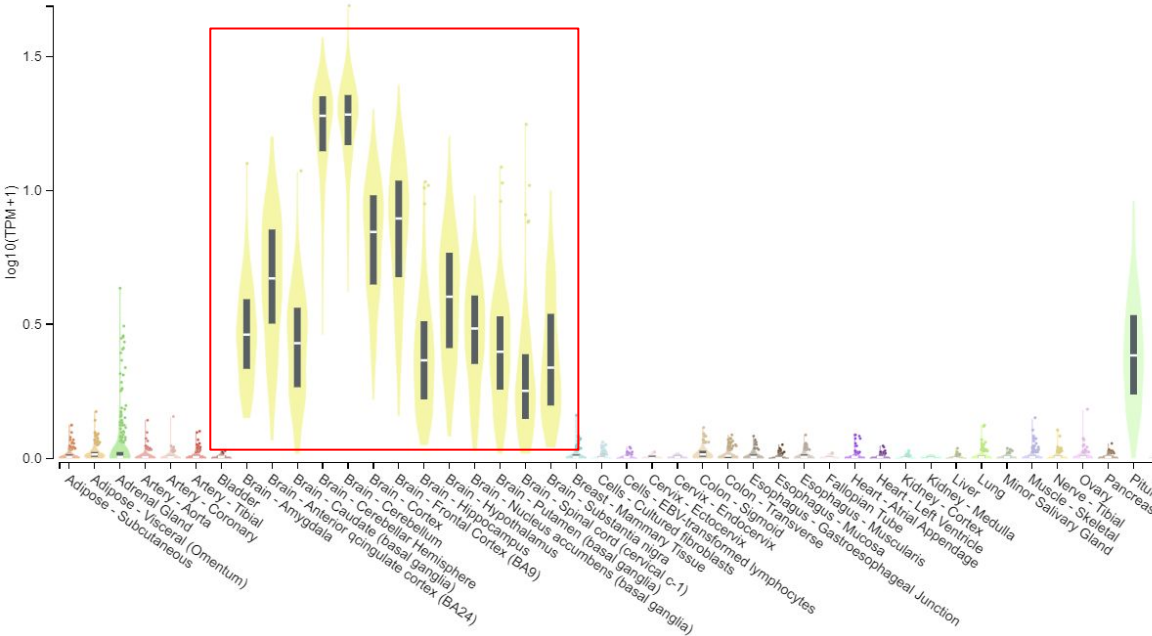
4) TBC1D22A



- **GTEx tissue expression:** broad expression among all tissue types, with highest levels in EBV-transformed lymphocytes, spleen, and the thyroid
- TBC1D22A is a GTPase activating protein of Rab40 which regulates lipophagy
- Loss of TBC1D22A results in lipid droplet accumulation across various tissues

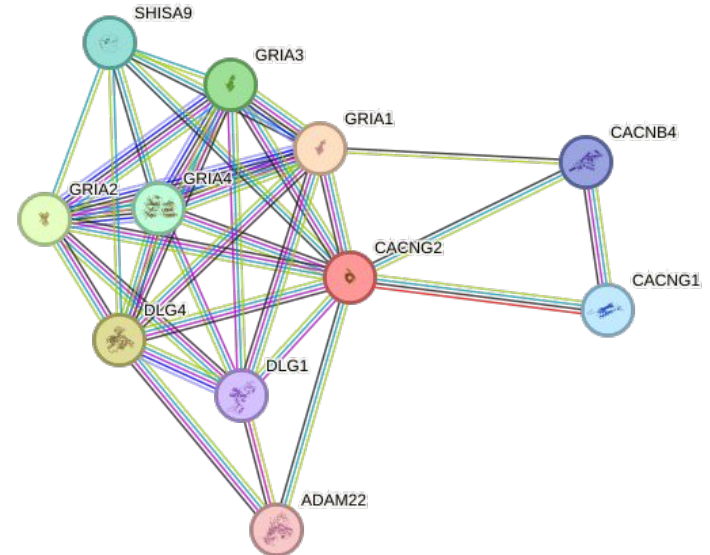
5) CACNG2

Bulk tissue gene expression for CACNG2 (ENSG00000166862.7)



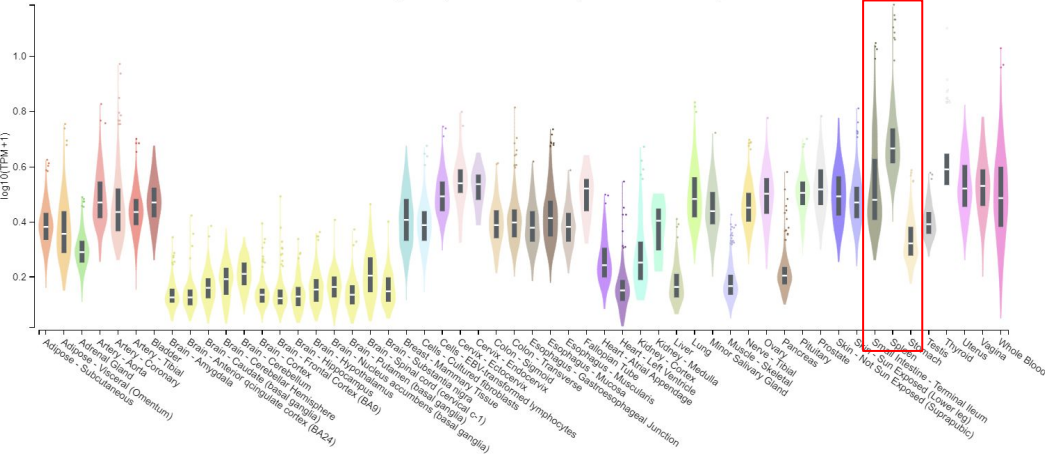
CACNG2 thus play an important role in synaptic plasticity, learning and memory. Disruptive mutations in this gene cause an autosomal dominant form of cognitive disability

Codes for voltage-dependent calcium channel gamma-2 subunit; Regulates the trafficking and gating properties of AMPA-selective glutamate receptors (AMPA-Rs).



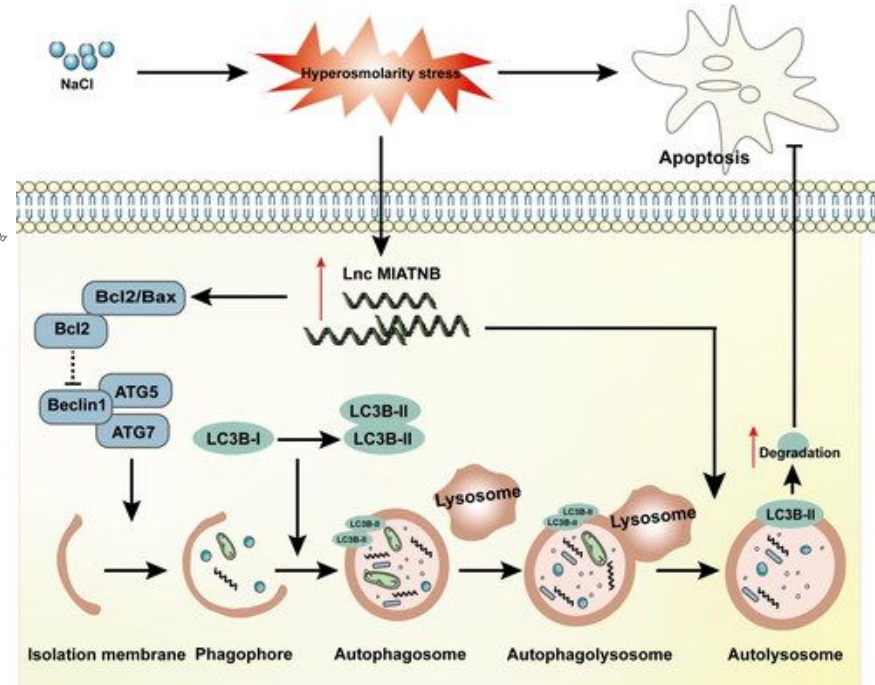
6) MIATNB

Bulk tissue gene expression for MIATNB (ENSG00000244625.7)

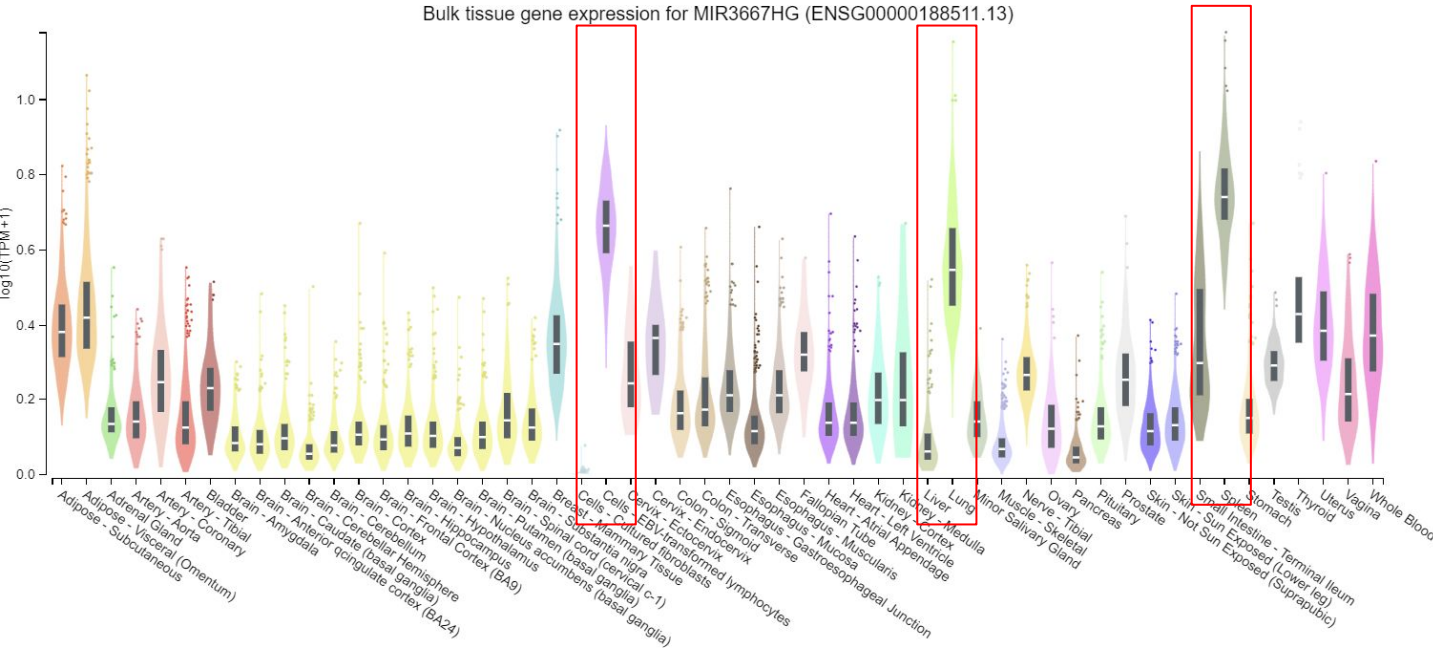


- Involved in hyperosmolarity-induced autophagy and apoptosis in human corneal epithelial cell (HCEC)-based model of dry eye disease.
- Hyperosmolarity upregulated MIATNB expression, while MIATNB knockdown inhibited autophagosome degradation

Myocardial infarction-associated transcript neighbor (MIATNB) is a long non-coding RNA that is most highly expressed in the spleen



7) MIR3667HG

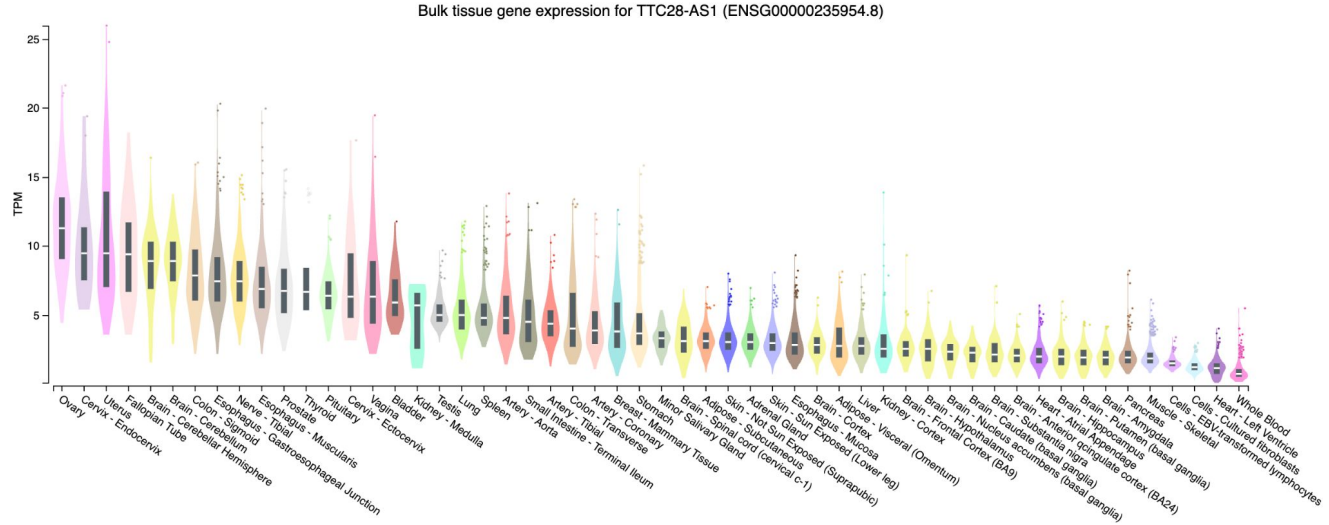


Highest expression in spleen, lung, and EBV-transformed lymphocytes.

Identified in extracellular vesicle (EV) DNA from pancreatic cancer patient-derived cells.

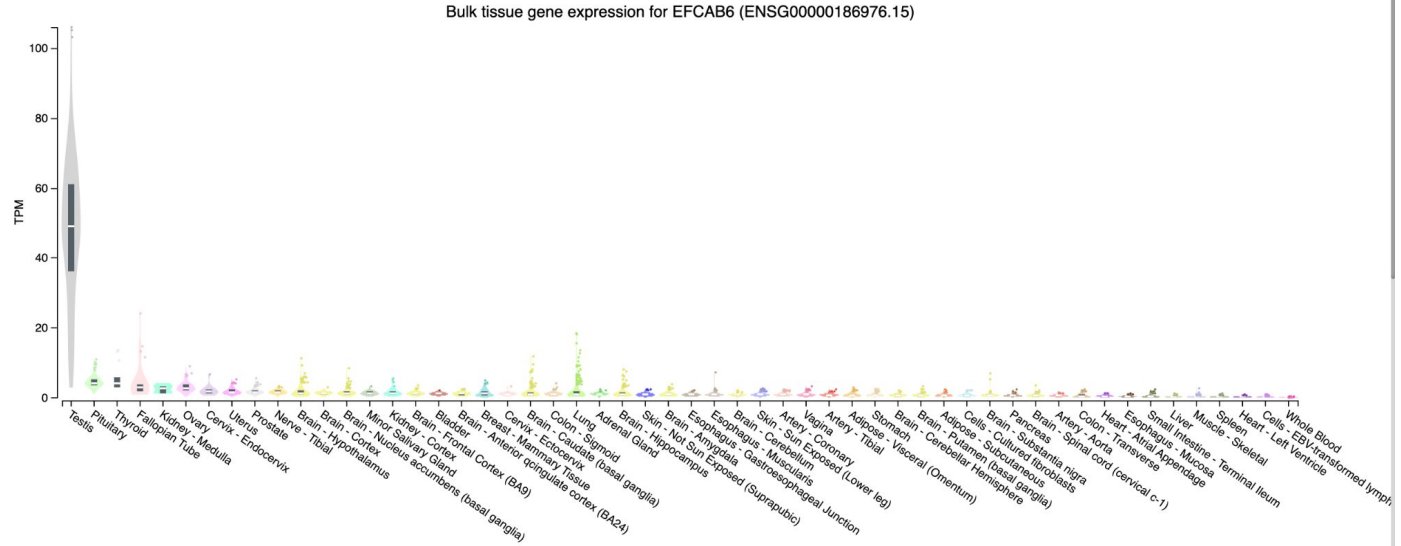
Low-molecular-weight DNA fragments not only distinguish cancer cells from normal cells but also exhibit cancer-specific enrichment of unique coding regions, including FAM135B, COL22A1, TSNARE1, KCNK9, ZFAT, JRK, MROH5, GSDMD, and **MIR3667HG** (Olou 2025)

8) TTC28



- **GTEx tissue expression:** primarily in the female reproductive system (ovaries, cervix, uterus), but it is expressed in many other tissues and organs as well
- Role in regulating mitosis and cytokinesis
- Contributes to chromosomal stability
- Downregulation and mutations are linked to several types of cancer

9) EFCAB6



- **GTEx tissue expression:** almost exclusively expressed in the testes
- Controls CatSper channel complexes (calcium channel responding to vaginal fluid pH)
- Role in regulating sperm motility and fertility
- Lack of expression contributes to CAIS (having XY chromosomes, but developing female genitalia)

Bulk tissue gene expression for LINC02885 (ENSG00000286592.1)

TPM

Brain - Hypothalamus
Brain - Frontal Cortex (BA9)
Brain - Anterior cingulate cortex (BA24)
Testis
Brain - Cortex
Brain - Hippocampus
Skin - Sun Exposed (Lower leg)
Skin - Substantia nigra
Skin - Not Sun Exposed (Suprapubic)
Brain - Cerebellar Hemisphere
Vagina
Cells - EBV-transformed lymphocytes
Pituitary
Brain - Nucleus accumbens (basal ganglia)
Brain - Caudate (basal ganglia)
Prostate
Thyroid
Ovary
Kidney - Cortex
Small Intestine - Terminal Ileum
Colon - Sigmoid
Colon - Transverse
Brain - Putamen (basal ganglia)
Minor Salivary Gland
Brain - Cerebellum
Cervix
Pancreas
Esophagus - Muscularis
Esophagus - Gastroesophageal Junction
Adipose - Visceral (Omentum)
Adipose - Subcutaneous
Artery - Aorta
Artery - Coronary
Bladder
Brain - Cerebellum
Breast - Mammary Tissue
Cells - Cultured fibroblasts
Esophagus - Mucosa
Esophagus - Esophagus
Fallopian Tube
Heart - Atrial Appendage
Heart - Left Ventricle
Liver
Lung
Muscle - Skeletal
Nerve - Tibial
Spleen
Stomach
Uterus
Whole Blood

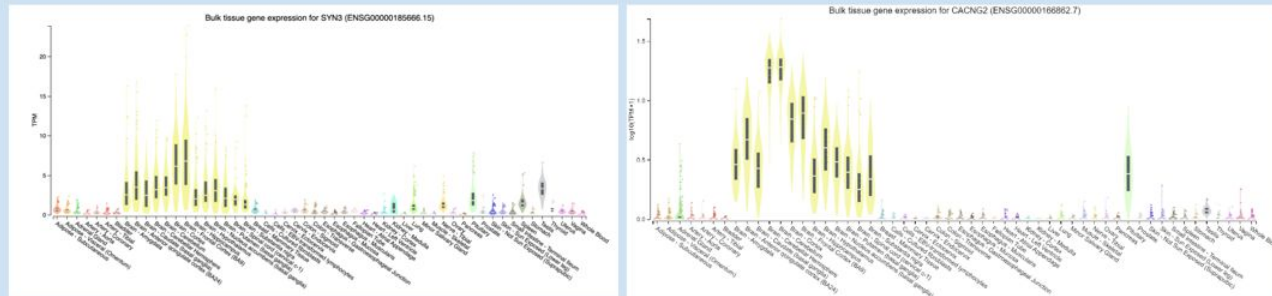
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2025 group 6 (chr 22)

Top 10 Prioritized Genes

1. LARGE1
2. SYN3
3. TAF A5
4. TBC1D22A
5. MIATNB
6. MIR3667HG
7. CACNG2
8. TTC28
9. EFCAB6
10. LINC02885

Summary Figure:



Summary:

1. Prioritization approach: highest mutational burden
2. Downstream analysis: tissue gene expression analysis
3. Findings:
 - a. 4 of 10 genes are highly expressed in brain tissues – TAF A5, SYN3, CACNG2, and LINC02885. Loss of these genes or genetic variants can lead to certain brain or mood disorders
 - b. 3 genes are linked to cancer - TTC28 (multiple cancer types), LINC02885 (gastric cancer), MIR3667HG (potential biomarker for malignant tumors)